

ETHAN VILLALOVOZ

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EDUCATION

Georgia Institute of Technology

Jan 2026 – May 2028

Master of Science in Computer Science — GPA: 4.0/4.0

Atlanta, GA

- Relevant Coursework: Robotics, Computer Graphics, Machine Learning, Natural Language Processing

Washington State University

Aug 2021 – May 2025

Bachelor of Science in Computer Science — GPA: 3.94/4.0

Pullman, WA

- Senior Design Project: [Retrieval-Augmented Generation Application Using Knowledge Graph and Vector Search](#)

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, C#, MATLAB, R, Haskell, Swift, GLSL

Developer Tools: Git, GitHub, GitHub Actions, Xcode, Docker, Bash, Conda, AWS, Postman, Jupyter, MLflow, MySQL

Libraries/Frameworks: React, Next.js, FastAPI, PyTorch, Hugging Face Transformers, OpenCV, WebGL, ROS

EXPERIENCE

Georgia Institute of Technology

Feb 2026 – Present

Graduate Student Researcher

Atlanta, GA

- Designing a **latent video-to-action framework** for robot policy learning using flow matching and diffusion

Microsoft

May 2026 – Jul 2026

Software Engineer Intern

Redmond, WA

- Developing a secure **multi-agent workflow system** for an internal cloud platform, focused on workflow integration

Washington State University

Jan 2024 – May 2025

Undergraduate Research Assistant

Pullman, WA

- Executed large-scale empirical evaluations of a **Bayesian Optimization**-based **prompt search** method for LLM-driven code generation, measuring functional correctness across **164 HumanEval+ tasks** and multiple models
- Built and ran reproducible experiment pipelines for **test-driven code synthesis**, benchmarking prompt optimization against **Chain-of-Thought** and **OPRO** baselines using pass@1 accuracy across HumanEval+ tasks
- Analyzed performance trends across optimization iterations, demonstrating **sample-efficient improvements** in code correctness through continuous embedding-space search with **Gaussian Process** surrogate models

Carnegie Mellon University

Jun 2024 – Aug 2024

Robotics Institute Summer Scholar

Pittsburgh, PA

- Developed a novel hierarchical **reward learning framework** using **Bayesian inference** to align robotic actions with human preferences from iterative **state corrections**, significantly enhancing robot adaptability
- Implemented a **proactive clarification dialogue** system that improved task accuracy by **30%** by resolving uncertainty through targeted human queries, reducing errors and advancing interactive human-robot collaboration
- Engineered a modular, extensible **Python**-based simulation environment using **Markov Decision Processes (MDP)**, supporting robust evaluation and iterative development of learning algorithms in simulated robotics tasks

Google

May 2023 – Aug 2023

Software Engineering Intern (STEP)

Sunnyvale, CA

- Developed and deployed **5 C++ and SQL**-based analytics jobs for internal database queue metrics, significantly reducing operational costs and enabling data-driven decision-making in collaboration with engineering stakeholders
- Optimized data sampling strategies to scale job execution from **1%** to **100%** dataset coverage within **4 hours**, achieving a **66%** reduction in runtime and improving the scalability, accuracy, and efficiency of internal analytics workflows
- Built interactive, real-time dashboards using **HTML** and **SQL**-based queries, delivering actionable insights to internal teams across engineering and operations, and enabling faster decision-making through intuitive visualizations
- Implemented live-update statistical features on client dashboards with **HTML** and database-driven queries, enhancing stakeholder visibility into queue activity, reducing detection latency, and enabling more responsive system oversight

PUBLICATIONS

[An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models](#). S. Tomar, A. Deshwal, [E. Villalovoz](#), M. Fazzini, H. Cai, J.R. Doppa. *arXiv*, 2025.

[Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach](#). A. Bacula, [E. Villalovoz](#), D. Flynn, A. Mehta, H. Knight. *International Conference on Intelligent Robots and Systems (IROS)*, 2023.